Question	Answer	Marks	Partial Marks
7(b)(ii)	[x =] 5.2[0] or 5.196… [y =] 6	**4**	**B2** or **M1** for $4\pi x^2 = 108\pi$ seen **B2** or **M1** for $\frac{1}{2}(4\pi y^2) + \pi y^2$ or better seen
8(a)(i)	× ✓ ✓ × × ✓	**4**	**B3** for 5 correct **B2** for 4 correct **B1** for 3 correct
8(a)(ii)	$\begin{pmatrix} 5 \\ 3 \end{pmatrix}$	**1**	Fraction line and/or missing brackets scores 0
8(a)(iii)	$\begin{pmatrix} 4 & 8 \\ 1 & 2 \end{pmatrix}$	**2**	**B1** for 2 or 3 correct elements (dep on 2 × 2 matrix)
8(a)(iv)	$\frac{1}{2}\begin{pmatrix} 3 & -1 \\ -4 & 2 \end{pmatrix}$ oe isw	**2**	**B1** for $k\begin{pmatrix} 3 & -1 \\ -4 & 2 \end{pmatrix}$ or determinant = 2 soi
8(b)	Rotation Origin oe 90 [anticlockwise] oe	**3**	**B1** for each
9(a)	$y = -2x + 5$ oe	**3**	**B2** for $-2x + 5$ or **M1** for gradient = $-1 \div \frac{1}{2}$ or better **M1** for substituting (1, 3) into y = (*their m*)$x + c$ oe If 0 scored **SC1** for (1, 3) satisfying their wrong equation ($c \neq 0$) with gradient $\neq \frac{1}{2}$
9(b)(i)	$x \geqslant 2$ oe $y \leqslant 5$ oe $y \geqslant \frac{1}{2}x$ oe	**4**	**SC3** for $x > 2$ and $y < 5$ and $y > \frac{1}{2}x$ OR **B1** for $x \geqslant 2$ **B1** for $y \leqslant 5$ **B2** for $y \geqslant \frac{1}{2}x$ or **M1** for $y \geqslant kx$ ($k > 0$) OR **SC2** for all three boundary lines identified but with incorrect sign(s) If 0 scored **SC1** for one or two correct boundary lines with incorrect sign(s)

Question	Answer	Marks	Partial Marks
9(b)(ii)	(5, 4)	**2**	**M1** for one trial of an integer point inside region or for $3x+5y=35$ drawn
10(a)(i)	26	**2**	**M1** for g(5) or for $\left(x^2+1\right)^2+1$
10(a)(ii)	x^2+4x+5	**2**	**M1** for $(x+2)^2+1$
10(a)(iii)	5	**2**	**M1** for $2x-3=7$
10(a)(iv)	$\frac{x+3}{2}$ oe	**2**	**M1** for $x=2y-3$ or $y+3=2x$ or $\frac{y}{2}=x-\frac{3}{2}$ oe
10(b)(i)	[0].70 cao	**2**	**B1** for [0].696 to [0].697
10(b)(ii)	4 cao	**1**	